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DAY 4

Task 1

Pose Detection Accuracy Improvement Documentation

Objective

The main objective of this task was to improve the accuracy, stability, and real-time performance of human pose detection using AI-based computer vision techniques. The project was implemented using MediaPipe and OpenCV in Python.

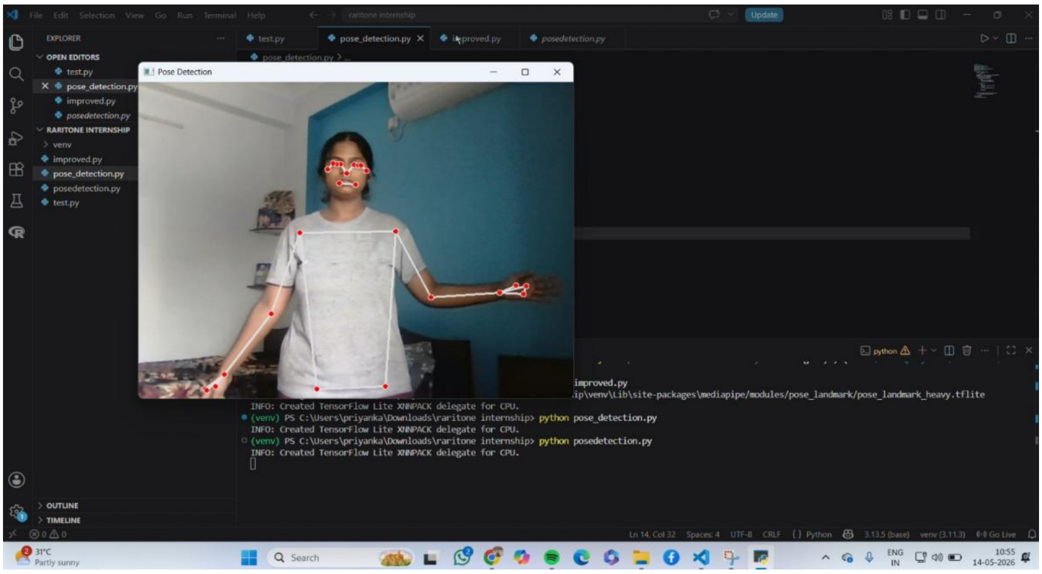
Initial Pose Detection Result

In the initial implementation:

- Basic pose landmarks were detected successfully.
- Skeleton tracking worked for body posture and arm movement.
- However, some landmark points appeared slightly unstable.
- Minor jitter and inconsistent hand tracking were observed during movement.
- Detection quality was affected by lighting and camera positioning.

Observations

- Lower smoothness in body landmark tracking
- Slight flickering in arm and hand points
- Reduced detection confidence
- Less stable skeleton structure

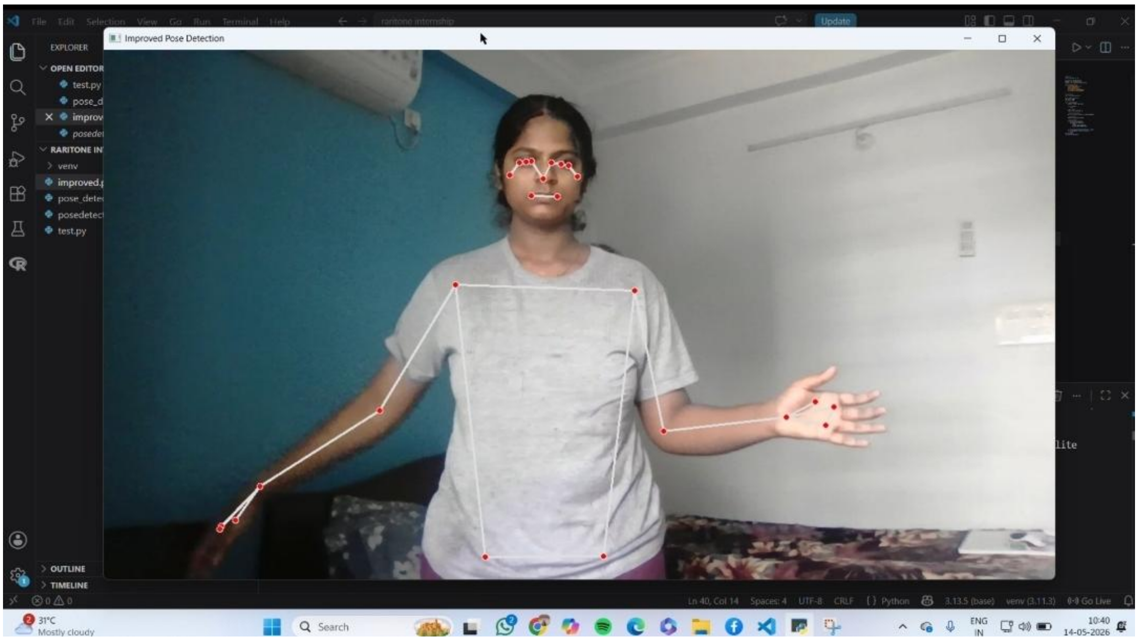


Improved Pose Detection Result

After implementing multiple optimizations, the pose detection system showed significant improvement.

Improvements Applied

- Increased detection confidence threshold
- Increased tracking confidence
- Enabled landmark smoothing
- Improved camera positioning
- Used higher webcam resolution
- Optimized body visibility inside the frame



Comparison Between Initial and Improved Results

Feature	Initial Detection	Improved Detection
Landmark Stability	Moderate	Highly Stable
Hand Tracking	Slightly Inaccurate	More Accurate
Skeleton Smoothness	Basic	Smooth & Refined
Detection Confidence	Lower	Higher
Flickering/Jitter	Present	Reduced
Real-Time Tracking	Average	Improved
Body Alignment	Moderate	Better Alignment

Detailed Improvements Observed

1. Better Landmark Accuracy

The improved model detected body joints more precisely, especially around:

- Arms
- Shoulders
- Hands
- Facial landmarks

The skeleton structure appeared more aligned with the actual body posture.

2. Reduced Jitter and Flickering

By enabling landmark smoothing and increasing tracking confidence:

- Sudden shaking of landmark points was minimized.
- Continuous motion tracking became smoother.
- Real-time movement appeared more natural.

3. Improved Hand and Arm Detection

The improved implementation handled arm extension and hand movement more accurately compared to the initial version.

This resulted in:

- Better joint positioning
- Improved tracking consistency
- More reliable body movement analysis

4. Enhanced Real-Time Performance

The optimized pose detection system processed body landmarks more efficiently during live webcam capture.

Benefits:

- Faster tracking response
- Stable frame processing
- Better user interaction experience

Optimization Techniques Used

Confidence Threshold Tuning

Higher detection and tracking confidence values improved the reliability of landmark prediction.

```
min_detection_confidence = 0.7

min_tracking_confidence = 0.7
```

Landmark Smoothing

Smoothing reduced sudden landmark fluctuations.

```
smooth_landmarks = True
```

Higher Resolution Processing

Higher camera resolution improved body feature visibility and landmark precision.

```
cap.set(3, 1280)

cap.set(4, 720)
```

Final Outcome

The improved pose detection system achieved: Better tracking accuracy, Stable landmark detection, Reduced flickering, Improved hand and body alignment, Enhanced real-time performance

The final implementation provided a more reliable foundation for future AI applications such as: Virtual Try-On Systems, Fitness Tracking, Gesture Recognition, Body Measurement Estimation, Clothing Recommendation Systems

Conclusion

The pose detection enhancement task successfully improved the quality and stability of real-time human pose estimation. By optimizing confidence thresholds, smoothing landmarks, and improving input quality, the system achieved more accurate and refined body tracking results suitable for advanced AI/ML applications.